SEMTM0051 — Foundations of Artificial Intelligence

Linear classification

Instructor: Roberto Colomboni version of 13 October 2026

4 Exercises

The exercise numbering is the same as in the lecture notes. References to earlier sections and equations refer to those notes.

The exercises follow the same path as the chapter. We first practise the sigmoid derivative in Exercise 1, and then compare logistic loss with the number of mistakes in Exercise 2. Exercise 3 works through one binary gradient-descent update, while Exercise 4 practises softmax and multiclass cross-entropy.

Throughout the exercises, use natural logarithms and sums of losses, without dividing by N, unless a mean is explicitly specified. In particular, remember the factor $1/\log 2$ when using (20). A calculator may be used for the final numerical values, but keep exact expressions until that final step.

1. **Derivative of the logistic sigmoid.** Starting from $\sigma(s) = (1 + e^{-s})^{-1}$, use the chain rule to show that

$$\sigma'(s) = \sigma(s)\sigma(-s) = \sigma(s)(1 - \sigma(s)) \qquad \text{for every } s \in \mathbb{R}.$$

   To organise the calculation, first identify the inner and outer functions and find their derivatives. Then apply the chain rule, using (21) to obtain the final equality.

2. **A smaller surrogate loss need not mean fewer mistakes.** Two observations both have label 1. One probability predictor assigns class 1 probabilities $0.51, 0.51$, while another assigns $0.49, 0.99$. In both cases, turn these probabilities into labels using the threshold rule $\text{step}(p - 1/2)$.

   (a) Compute each classifier's predicted labels, number of mistakes, and sum of logistic losses $\ell_{\log}$.

   (b) Using your calculations, determine which classifier is preferred by each objective. Then explain why their different preferences do not contradict (20).

   (c) Now consider a dataset of N observations on which the mean logistic loss is at most 0.03. What error-rate bound follows from (20)? Separately, suppose only that the sum of logistic losses is at most $\log 2$. Does this guarantee zero classification mistakes for every dataset? Justify your answer with a proof or a counterexample.

3. One binary gradient-descent update. Consider the dataset

i	1	2	3
x_i	-1	0	2
y_i	0	1	1

and the probability model $p_{\boldsymbol{\theta}}(x) = \sigma(b + ax)$, with $\boldsymbol{\theta} = (b, a)^\mathsf{T}$. Start from

$$\boldsymbol{\theta}^{(0)} = \begin{pmatrix} 0 \\ \log 3 \end{pmatrix}, \qquad \eta = \frac{1}{5}.$$

(a) First construct $\boldsymbol{X}$, and then compute the score column $\boldsymbol{s}^{(0)}$, probability column $\boldsymbol{p}^{(0)}$, and differences $\boldsymbol{r}^{(0)} = \boldsymbol{p}^{(0)} - \boldsymbol{y}$. Keep these values exact.

(b) Using these differences, apply (35) to calculate the gradient. Then perform one simultaneous update using (37), giving both the new intercept and the new slope.

(c) Compute the predicted labels and empirical logistic loss before and after the update. Use the tie rule from (2). Determine the number of mistakes before and after the update. Find the decision threshold before and after the update, and identify which observation changes predicted class as the threshold moves.

4. Softmax and cross-entropy. Consider two observations with three classes. Their score rows are

$$\boldsymbol{s}_1 = (0, \log 3, \log 3), \qquad \boldsymbol{s}_2 = (\log 4, 0, \log 2),$$

and their observed labels are

$$y_1 = 3, \qquad y_2 = 1.$$

(a) Apply softmax to each score row. For each observation, first exponentiate the scores, then add the exponentials, and finally divide each exponential by their sum. Keep the resulting probabilities exact.

(b) Which class is predicted for each observation? Apply our rule for ties and state whether each prediction is correct.

(c) Compute the categorical cross-entropy loss of each observation,

$$\ell_{\text{multi}}(\text{softmax}(\boldsymbol{s}_i), y_i),$$

and hence compute the summed loss over the two observations.

(d) Verify the two individual losses again using the score form

$$\ell_{\text{multi}}(\text{softmax}(\boldsymbol{s}), y) = -s_y + \log\left(\sum_{k=1}^{K} e^{s_k}\right).$$

Solutions

1. Derivative of the logistic sigmoid.

To identify the two functions in the composition, write

$$u : \mathbb{R} \to (1, \infty), \quad u(s) = 1 + e^{-s}, \qquad v : (0, \infty) \to \mathbb{R}, \quad v(r) = r^{-1}.$$

Then $\sigma = v \circ u$, and the derivatives of the two functions are

$$u'(s) = -e^{-s}, \qquad v'(r) = -r^{-2}.$$

The minus sign in u' comes from applying the chain rule to e^{-s}. Therefore, applying the chain rule once more to $v \circ u$, we obtain

$$\begin{aligned} \sigma'(s) &= v'(u(s))u'(s) \\ &= -\left(1 + e^{-s}\right)^{-2} \left(-e^{-s}\right) \\ &= \frac{e^{-s}}{(1 + e^{-s})^2}. \end{aligned}$$

Since the two minus signs cancel, the result is positive. To express it in terms of the sigmoid, we now factor the fraction as

$$\sigma'(s) = \frac{1}{1 + e^{-s}} \frac{e^{-s}}{1 + e^{-s}}.$$

The first factor is $\sigma(s)$. To identify the second, multiply its numerator and denominator by e^s, which gives

$$\frac{e^{-s}}{1 + e^{-s}} = \frac{1}{1 + e^{s}} = \sigma(-s).$$

Therefore, for every real s,

$$\boxed{\sigma'(s) = \sigma(s)\sigma(-s) = \sigma(s)(1 - \sigma(s)).}$$

In particular, the derivative is positive for every finite score, consistent with the increasing sigmoid curve.

2. A smaller surrogate loss need not mean fewer mistakes.

(a) Predictions and losses. Both probabilities from the first predictor are above $1/2$, so its classifier predicts $(1, 1)$ and makes zero mistakes. Since both observed labels are 1, its summed logistic loss is

$$-2\log(0.51) \approx 1.34669.$$

By contrast, the second predictor assigns a probability below $1/2$ to the first observation. Therefore its classifier predicts $(0, 1)$ and makes one mistake, while its summed logistic loss is

$$-\log(0.49) - \log(0.99) \approx 0.72340.$$

(b) The objectives prefer different classifiers. These calculations show that the error count prefers the first classifier, while the logistic loss prefers the second. However, the error bound remains valid for both, since

$$0 \le \frac{-2\log(0.51)}{\log 2} \approx 1.94286, \qquad 1 \le \frac{-\log(0.49) - \log(0.99)}{\log 2} \approx 1.04365.$$

In fact, the upper bound compares the two quantities at the same predictor, so it does not require them to rank different predictors in the same order. Here the small probability decrease on the first observation changes its predicted label. However, the large probability increase on the second observation more than compensates for it in the surrogate loss. Therefore the total logistic loss decreases even though the number of mistakes increases. Multiplying both surrogate values by $1/\log 2$ would preserve this comparison.

(c) Guarantees from a small loss. Dividing the error-count bound by the number of observations shows that the error rate is at most the mean logistic loss divided by $\log 2$. Therefore, in this case, it is at most

$$\frac{0.03}{\log 2} \approx 0.04328,$$

or approximately 4.33%, on the same dataset.

The separate assumption that the summed logistic loss is at most $\log 2$ does not, in general, guarantee zero mistakes. Consider a dataset with one observation whose label is 1, and suppose that the predictor assigns probability $1/2$ to class 1. The summed loss is then just

$$\ell_{\log}(1/2, 1) = -\log(1/2) = \log 2.$$

However, the predicted label is

$$\operatorname{step}\left(\frac{1}{2} - \frac{1}{2}\right) = \operatorname{step}(0) = 0,$$

so the classifier makes one mistake. The error-count bound gives at most one mistake in this case, which is consistent with the example. By contrast, a summed loss strictly less than $\log 2$ gives an error-count bound strictly less than one. Since the number of mistakes is a nonnegative integer, the strict inequality guarantees zero mistakes.

3. One binary gradient-descent update.

(a) Scores, probabilities and differences. Write $p_i^{(t)} = p_{\boldsymbol{\theta}^{(t)}}(x_i)$. Adding the intercept column gives the design matrix; the labels form the column $\boldsymbol{y}$:

$$\boldsymbol{X} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{pmatrix}, \qquad \boldsymbol{y} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

At the initial parameters, the scores, probabilities and differences are

$$\boldsymbol{s}^{(0)} = \begin{pmatrix} -\log 3 \\ 0 \\ 2\log 3 \end{pmatrix}, \qquad \boldsymbol{p}^{(0)} = \begin{pmatrix} 1/4 \\ 1/2 \\ 9/10 \end{pmatrix}, \qquad \boldsymbol{r}^{(0)} = \begin{pmatrix} 1/4 \\ -1/2 \\ -1/10 \end{pmatrix}.$$

For instance, $e^{2\log 3} = 9$, so the third probability is $9/(1+9) = 9/10$.

(b) Gradient and update. The intercept derivative sums the differences, while the slope derivative sums them after multiplying by the inputs. Therefore

$$\nabla \mathcal{L}_{\log}(\boldsymbol{\theta}^{(0)}) = \boldsymbol{X}^{\top}\boldsymbol{r}^{(0)} = \begin{pmatrix} 1/4 - 1/2 - 1/10 \\ -1/4 + 0 - 2/10 \end{pmatrix} = \begin{pmatrix} -7/20 \\ -9/20 \end{pmatrix}.$$

Thus

$$\boldsymbol{\theta}^{(1)} = \begin{pmatrix} 0 \\ \log 3 \end{pmatrix} - \frac{1}{5}\begin{pmatrix} -7/20 \\ -9/20 \end{pmatrix} = \begin{pmatrix} 7/100 \\ \log 3 + 9/100 \end{pmatrix}.$$

Both entries use the same $\boldsymbol{r}^{(0)}$: we update the intercept and slope simultaneously before recomputing the probabilities.

(c) Predictions and losses. Initially, the predicted labels are $(0, 0, 1)$: the tie is assigned to class 0, so only the second observation is misclassified. After the update,

$$\boldsymbol{s}^{(1)} = \begin{pmatrix} -\log 3 - 0.02 \\ 0.07 \\ 2\log 3 + 0.25 \end{pmatrix}, \qquad \boldsymbol{p}^{(1)} \approx \begin{pmatrix} 0.246269 \\ 0.517493 \\ 0.920358 \end{pmatrix}.$$

The predicted labels become $(0, 1, 1)$, giving zero mistakes. The logistic loss also decreases, from

$$\mathcal{L}_{\log}(\boldsymbol{\theta}^{(0)}) = -\log(3/4) - \log(1/2) - \log(9/10) = \log(80/27) \approx 1.086190,$$

to the following value, calculated from the unrounded probabilities:

$$\mathcal{L}_{\log}(\boldsymbol{\theta}^{(1)}) = -\log(1 - p_1^{(1)}) - \log p_2^{(1)} - \log p_3^{(1)} \approx 1.024471.$$

The decision threshold solves $b + ax = 0$. Before and after the update, respectively, it is

$$-\frac{b^{(0)}}{a^{(0)}} = 0, \qquad -\frac{b^{(1)}}{a^{(1)}} = -\frac{7/100}{\log 3 + 9/100} \approx -0.0589.$$

Both slopes are positive, so inputs to the right of the threshold receive class 1, while inputs at or to its left receive class 0. Initially, the second observation, at $x_2 = 0$, lies exactly on the threshold and therefore receives class 0. After the update, the threshold has moved to the left of this observation, which now receives class 1 and is classified correctly. The inputs $x_1 = -1$ and $x_3 = 2$ stay on their original sides of the threshold, so their predicted labels remain unchanged.

4. Softmax and cross-entropy.

(a) Softmax probabilities. For the first observation, $\boldsymbol{s}_1 = (0, \log 3, \log 3)$. Exponentiating gives:

$$(e^0, e^{\log 3}, e^{\log 3}) = (1, 3, 3).$$

Their sum is $1 + 3 + 3 = 7$, so dividing each entry by this common sum gives

$$\text{softmax}(\boldsymbol{s}_1) = \left(\frac{1}{7}, \frac{3}{7}, \frac{3}{7}\right).$$

For the second observation, $\boldsymbol{s}_2 = (\log 4, 0, \log 2)$. Exponentiating gives

$$(e^{\log 4}, e^0, e^{\log 2}) = (4, 1, 2),$$

whose entries sum to 7. Therefore,

$$\text{softmax}(\boldsymbol{s}_2) = \left(\frac{4}{7}, \frac{1}{7}, \frac{2}{7}\right).$$

(b) Predicted classes. For the first observation, classes 2 and 3 share the largest probability, 3/7. Our rule chooses the smaller label, class 2. Since $y_1 = 3$, this prediction is incorrect.

For the second observation, the largest probability is 4/7, in the first position, so the classifier predicts class 1. Since $y_2 = 1$, this prediction is correct.

(c) Cross-entropy losses. Cross-entropy uses the probability assigned to the observed class. For the first observation, the observed class is 3 and receives probability 3/7. Hence

$$\ell_{\text{multi}}(\text{softmax}(\boldsymbol{s}_1), y_1) = -\log(3/7) = \log(7/3).$$

For the second observation, the observed class is 1 and receives probability 4/7. Hence

$$\ell_{\text{multi}}(\text{softmax}(\boldsymbol{s}_2), y_2) = -\log(4/7) = \log(7/4).$$

Therefore the summed loss is

$$\log(7/3) + \log(7/4) = \log(49/12) \approx 1.40691.$$

(d) Checking the score form. Write $\boldsymbol{s}_i = (s_{i1}, s_{i2}, s_{i3})$. For the first observation, $y_1 = 3$, so

$$\begin{aligned} -s_{1,y_1} + \log\left(\sum_{k=1}^{3} e^{s_{1k}}\right) &= -\log 3 + \log(1 + 3 + 3) \\ &= \log(7/3). \end{aligned}$$

For the second observation, $y_2 = 1$, so

$$\begin{aligned} -s_{2,y_2} + \log\left(\sum_{k=1}^{3} e^{s_{2k}}\right) &= -\log 4 + \log(4 + 1 + 2) \\ &= \log(7/4). \end{aligned}$$

These are exactly the same losses obtained from the probabilities.